

1 Diophantine Equations

1.1 Introduction and Fundamentals

Diophantine equations are polynomial equations whose solutions are restricted to integers. Named after the ancient mathematician Diophantus, these equations form a crucial part of number theory. Unlike regular equations where we accept any real or complex solutions, Diophantine equations specifically seek integer solutions.

1.2 Linear Diophantine Equations

The most basic form is the linear Diophantine equation:

$$ax + by = c$$

where a , b , and c are integers, and we seek integer solutions for x and y .

(Existence of Solutions) A linear Diophantine equation $ax + by = c$ has integer solutions if and only if the greatest common divisor of a and b divides c . In other words:

$$\text{Solutions exist} \iff \gcd(a, b) \mid c$$

(General Solution) If (x_0, y_0) is a particular solution, then all solutions are given by:

$$x = x_0 + \frac{b}{\gcd(a, b)}t, \quad y = y_0 - \frac{a}{\gcd(a, b)}t$$

where t ranges over all integers.

1.3 Methods of Solution

1. **Extended Euclidean Algorithm** Used to find particular solutions of $ax + by = \gcd(a, b)$
2. **Back-substitution** Once we have a solution for $\gcd(a, b)$, multiply by appropriate factor
3. **Modular Arithmetic** Often useful in solving systems of Diophantine equations

1.4 Special Types

1. **Pythagorean Equations:** $x^2 + y^2 = z^2$ General solution: $x = k(m^2 - n^2)$, $y = k(2mn)$, $z = k(m^2 + n^2)$
2. **Pell's Equation:** $x^2 - ny^2 = 1$ Solutions found using continued fractions

Practice Problems - Diophantine Equations

1. Find all integer solutions to $15x + 21y = 9$.
2. Solve in positive integers: $x^2 - 5y^2 = 4$.
3. Find all integer solutions to $3x + 5y = 7$ where $x \geq 0$.
4. Find all positive integer solutions to $x^2 + y^2 = 325$.

5. Find all positive integer solutions to $6x + 10y + 15z = 100$.

Solutions:

1. $\gcd(15, 21) = 3$ divides 9, so solutions exist. Using extended Euclidean: $1 = 3(5) - 2(7)$
Multiply by 3: $3 = 9(5) - 6(7)$ Therefore general solution: $x = 9 + 7t$, $y = -6 - 5t$ where $t \in \mathbb{Z}$
2. Factor as $(x + \sqrt{5}y)(x - \sqrt{5}y) = 4$ Factor 4 into $1 \cdot 4$, $2 \cdot 2$, $4 \cdot 1$ Solve system of equations for each case Only positive solution: $(x, y) = (3, 1)$
3. Using extended Euclidean: $1 = 2(3) - 1(5)$ Multiply by 7: $7 = 14(3) - 7(5)$ General solution: $x = 14 - 5t$, $y = -7 + 3t$ For $x \geq 0$: $14 - 5t \geq 0$ Therefore $t \leq 2$ Integer solutions: $(x, y) = \{(14, -7), (9, -4), (4, -1)\}$
4. Factor $325 = 13 \cdot 25$ Use fact that sum of squares formula: $(a^2 + b^2)(c^2 + d^2) = (ac \mp bd)^2 + (ad \pm bc)^2$ $325 = 13 \cdot 25 = (2^2 + 3^2)(5^2 + 0^2)$ Positive solutions: $(x, y) = (10, 15)$ and $(x, y) = (15, 10)$
5. Divide throughout by $\gcd(6, 10, 15) = 1$ Simplify: $6x + 10y + 15z = 100$ First solve for z : $z = \frac{100 - 6x - 10y}{15}$ For integer z : $100 - 6x - 10y$ must be divisible by 15 Try values: $x, y \geq 0$ and check when z is positive Solutions: $(5, 4, 2)$, $(8, 1, 3)$, $(2, 7, 1)$, $(11, 2, 0)$

2 Combinatorial Identities

2.1 Key Concepts

(Binomial Coefficient) The binomial coefficient $\binom{n}{k}$ represents the number of ways to choose k items from n items:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

(Pascal's Identity) For positive integers n and k :

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

Proof. This identity has both algebraic and combinatorial proofs:

1. Algebraic Proof:

$$\begin{aligned} \binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!} \\ &= \frac{n!k + n!(n-k+1)}{k!(n-k+1)!} \\ &= \frac{(n+1)!}{k!(n+1-k)!} \\ &= \binom{n+1}{k} \end{aligned}$$

2. Combinatorial Proof:

Consider choosing k items from $n+1$ items. The first item can either be:

- Selected: We then need to choose $k - 1$ items from remaining n items
- Not selected: We then need to choose k items from remaining n items

(Row Sum) For any non-negative integer n :

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

Proof. This represents the total number of subsets of an n -element set, or equivalently:

$$(1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{n-k} = \sum_{k=0}^n \binom{n}{k}$$

2.2 Advanced Identities

(Hockey Stick Identity) For non-negative integers n and r where $r \leq n$:

$$\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$$

(Vandermonde's Identity) For non-negative integers m , n , and r :

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

2.3 Combinatorial Applications

Example: Committee Selection

From a group of 10 people, we need to form a committee of 4. If two specific people must be on the committee, how many ways can this be done?

Solution: We must: 1. Include the 2 required people 2. Select 2 more from the remaining 8 people Therefore: $\binom{8}{2} = 28$ ways

Example: Path Counting

Find the number of shortest paths from $(0, 0)$ to (n, k) moving only right and up.

Solution: Total steps needed: n right + k up = $n + k$ total steps We need to choose positions for either right or up moves Answer: $\binom{n+k}{n} = \binom{n+k}{k}$

Practice Problems - Combinatorial Identities

1. Prove that $\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}$.
2. Prove that $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.
3. Prove that $\binom{n}{r} \binom{r}{k} = \binom{n}{k} \binom{n-k}{r-k}$.
4. Show that $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$.
5. Prove that $\sum_{k=0}^r \binom{n+k}{k} = \binom{n+r+1}{r}$.

Solutions:

1. Use differentiation of $(1+x)^n$ $\frac{d}{dx}(1+x)^n = n(1+x)^{n-1}$ Expand both sides and compare coefficients Set $x = 1$ to get required result
2. Apply Vandermonde's Identity with $m = n$, $n = n$, $r = n$ Use symmetry property $\binom{n}{n-k} = \binom{n}{k}$ Direct substitution gives result
3. Write out each binomial coefficient using factorials $\frac{n!}{r!(n-r)!} \cdot \frac{r!}{k!(r-k)!} = \frac{n!}{k!(n-k)!} \cdot \frac{(n-k)!}{(r-k)!(n-r)!}$
Cancel common factors Both sides equal $\frac{n!}{k!(r-k)!(n-r)!}$
4. Use binomial theorem: $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$ Put $x = -1$ $(1-1)^n = \sum_{k=0}^n (-1)^k \binom{n}{k}$
 $0 = \sum_{k=0}^n (-1)^k \binom{n}{k}$
5. Use induction on r Base case: $r = 0$ is trivial Inductive step: Assume true for $r-1$ Use Pascal's identity on RHS Algebraic manipulation gives result

3 Trigonometry

3.1 Fundamental Concepts

(Basic Trigonometric Functions) For an angle θ in a right triangle:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\text{opposite}}{\text{adjacent}}$$

(Pythagorean Identity) For any angle θ :

$$\sin^2 \theta + \cos^2 \theta = 1$$

3.2 Advanced Identities

(Addition Formulas) For angles α and β :

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

(Double Angle Formulas)

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

3.3 Sum-to-Product Formulas

(Sum and Difference to Product)

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

3.4 Trigonometric Equations

(General Solutions) For any integer n :

$$\sin \theta = \sin \alpha \implies \theta = \alpha + 2\pi n \text{ or } \theta = \pi - \alpha + 2\pi n$$

$$\cos \theta = \cos \alpha \implies \theta = \pm \alpha + 2\pi n$$

Practice Problems - Trigonometry

1. Solve the equation: $2 \sin^2 x - 3 \sin x - 2 = 0$ in the interval $[0, 2\pi]$.
2. Prove: $\tan^2 x + 1 = \sec^2 x$
3. If $\sin A + \sin B = 1$ and $\cos A + \cos B = 1$, find $\sin(A+B)$.
4. Solve: $\cos 2x = \cos x$ in $[0, 2\pi]$
5. If $\tan A + \tan B = 1$ and $\tan A \tan B = 1$, find all possible values of $A+B$.

Solutions:

1. Let $y = \sin x$ $2y^2 - 3y - 2 = 0$ $(2y+2)(y-2) = 0$ $y = -1$ or $y = 2$ Since $|\sin x| \leq 1$, only $y = -1$ is valid Therefore $x = \frac{3\pi}{2}$ in $[0, 2\pi]$
2. $\tan^2 x + 1 = \frac{\sin^2 x}{\cos^2 x} + 1 = \frac{\sin^2 x + \cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$
3. Square and add both equations: $2 + 2 \sin A \sin B = 1$ $2 + 2 \cos A \cos B = 1$ Add these equations: $4 + 2(\sin A \sin B + \cos A \cos B) = 2$ $\sin A \sin B + \cos A \cos B = -1$ Therefore $\sin(A+B) = 0$
4. Use $\cos 2x = 2 \cos^2 x - 1$ $2 \cos^2 x - 1 = \cos x$ $2 \cos^2 x - \cos x - 1 = 0$ $(2 \cos x + 1)(\cos x - 1) = 0$ $\cos x = -\frac{1}{2}$ or $\cos x = 1$ $x = \frac{2\pi}{3}, \frac{4\pi}{3}, 0, 2\pi$
5. Let $\tan A = t$, then $\tan B = \frac{1}{t}$ From first equation: $t + \frac{1}{t} = 1$ $t^2 - t + 1 = 0$ $t = \frac{1 \pm \sqrt{-3}}{2}$ Therefore $A+B = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$

4 Complex Numbers

4.1 Basic Definitions

(Complex Number) A complex number z is an expression of the form:

$$z = a + bi$$

where a, b are real numbers and $i^2 = -1$. Here:

- a is the real part: $\Re(z)$
- b is the imaginary part: $\Im(z)$
- If $z = a + bi$, then its conjugate is $\bar{z} = a - bi$

4.2 Polar Form

(Polar Representation) Any complex number $z = a + bi$ can be written as:

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

where:

$$r = \sqrt{a^2 + b^2} \text{ (modulus)}$$

$$\theta = \arctan\left(\frac{b}{a}\right) \text{ (argument)}$$

(De Moivre's Formula) For any real number θ and integer n :

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

4.3 Complex Operations

(Basic Operations) For complex numbers $z_1 = a + bi$ and $z_2 = c + di$:

$$z_1 + z_2 = (a + c) + (b + d)i$$

$$z_1 \cdot z_2 = (ac - bd) + (ad + bc)i$$

$$\frac{z_1}{z_2} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2} \text{ (when } z_2 \neq 0)$$

4.4 Properties of Complex Numbers

(Properties of Conjugates) For complex numbers z and w :

$$\overline{z + w} = \bar{z} + \bar{w}$$

$$\overline{z \cdot w} = \bar{z} \cdot \bar{w}$$

$$z \cdot \bar{z} = |z|^2 = (\Re(z))^2 + (\Im(z))^2$$

(Roots of Unity) The n th roots of unity are the complex numbers satisfying $z^n = 1$:

$$z_k = \cos\left(\frac{2\pi k}{n}\right) + i \sin\left(\frac{2\pi k}{n}\right), \quad k = 0, 1, \dots, n-1$$

4.5 Applications

(Euler's Formula) For any real number x :

$$e^{ix} = \cos x + i \sin x$$

(Complex Exponentials) For complex numbers:

$$e^{z_1} \cdot e^{z_2} = e^{z_1+z_2}$$

$$(e^z)^n = e^{nz}$$

4.6 Important Corollaries

(Trigonometric Identities from Complex Numbers) Using Euler's formula:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

Practice Problems - Complex Numbers

1. Find all complex numbers z such that $|z+1| = |z-i|$.
2. Find all complex numbers z such that $z^4 = 16i$.
3. If $z = \frac{1+i}{1-i}$, find z^2 .
4. Find all complex numbers z such that $z\bar{z} + z + \bar{z} = 3$.
5. Find all complex numbers z such that $|z| + |z-1| = 2$.

Solutions:

1. Using definition of complex modulus: $(x+1)^2 + y^2 = x^2 + (y-1)^2$ Expand and simplify:
 $2x+1 = -2y+1$ Therefore $x = -y$ Answer: Points on line $x+y=0$
2. $16i = 16e^{i\pi/2}$ $z = \sqrt[4]{16}e^{i(\pi/8+k\pi/2)}$ $z = 2e^{i(\pi/8+k\pi/2)}$, $k = 0, 1, 2, 3$ Four solutions: $2e^{i\pi/8}$, $2e^{5i\pi/8}$, $2e^{9i\pi/8}$, $2e^{13i\pi/8}$
3. First multiply numerator and denominator by conjugate of denominator: $z = \frac{(1+i)(1+i)}{(1-i)(1+i)} = \frac{1+2i-1}{1+1} = i$ Therefore $z^2 = i^2 = -1$
4. Let $z = x+yi$ $z\bar{z} = x^2 + y^2$ $z + \bar{z} = 2x$ Therefore: $x^2 + y^2 + 2x = 3$ $(x+1)^2 + y^2 = 4$ Circle with center $(-1,0)$ and radius 2
5. This is the definition of an ellipse Sum of distances from focus points $(0,0)$ and $(1,0)$ equals 2 Therefore points lie on ellipse with foci at 0 and 1. Major axis = 2